(e) $F(x,y,z) = 1$

Sum of minterms: $\Sigma(0,1,2,3,4,5,6,7)$

No product of maxterms

(f) $F(x,y,z) = (xy+z)(y+xz)$

$= (z+xy)(y+xz)$ (Postulate 3(a))

$= (z+x)(z+y)(y+x)(y+z)$ (Postulate 4(b))

$= ((z+x)+0)((z+y)+0)((y+x)+0)((y+z)+0)$ (Postulate 2(a))

$= ((z+x)+yy')((z+y)+xx')((y+x)+zz')((y+z)+xx')$ (Postulate 5(b))

$= ((z+x)+yy')((y+z)+xx')+((x+y)+zz')((y+z)+xx')$

(Postulate 3(a))

$= ((z+x)+y)((z+x)+y')((y+z)+x)((y+z)+x')((x+y)+z)((x+y)+z')$

$((y+z)+x)((y+z)+x')$ (Postulate 4(b))

$= (z+(x+y))(z+(x+y'))((y+z)+x)((y+z)+x')((x+y)+z)((x+y)+z')$

$((y+z)+x)((y+z)+x')$ (Theorem 4(a))

$= ((x+y)+z)((x+y')+z)(x+(y+z))(x'+(y+z))((x+y)+z)((x+y)+z')$

$(x+(y+z))(x'+(y+z))$ (Postulate 3(a))

$= (x+y+z)(x+y'+z)(x+y+z)(x'+y+z)(x+y+z)(x+y+z')(x+y+z)(x'+y+z)$

(Theorem 4(a))

Removing all repetitive terms, we have:

$(x+y+z)(x+y+z')(x+y'+z)(x'+y+z)$

$\Pi(0,1,2,4)$ $\Sigma(3,5,6,7)$

$\boxed{2\text{-}18:}$ Show that the dual of the exclusive-OR is equal to its complement.

We know that the algebraic expression of the exclusive-OR is $xy' + x'y$.

Dual: $(x+y')\cdot(x'+y)$

$(xy'+x'y)' = (xy')'\cdot(x'y)'$ (Theorem 5(a))

$= (x'+(y')')\cdot((x')'+y')$ (Theorem 5(b))

$= (x'+y)\cdot(x+y')$ (Theorem 3)